GOAT_666 Online Travel Platform

Software Requirements Specification (SRS)

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Note: This document was prepared based on the Project Definition and the course assignment requirements, following the requirements expression principles of GB/T 9385-2008, Specification for Computer Software Requirements, and GB/T 45802-2025, Systems and Software Engineering — Life Cycle Processes — Requirements Engineering. Functional requirements FR-01 – FR-12, scenarios SC-01 – SC-06, pages UI-01 – UI-08, and acceptance tests AT-F01 – AT-F10 are referenced consistently throughout. All initial release functions have verifiable requirements; additional functions are marked "Extension." Unfinished items are recorded in Chapter 6.

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1. Introduction

1.1 Purpose

This document defines the business requirements, user behaviors, functional rules, exception handling logic, and acceptance criteria for the GOAT_666 online travel platform. For the development team, this specification serves as the requirements baseline for system design and coding, ensuring that all members maintain a consistent understanding of the product logic. For testers, this document provides the basis for writing test cases and conducting functional and interface validation. For course review instructors, this document constitutes a complete requirements deliverable for the software engineering course assignment. This specification is intended solely for the course practice project and shall not serve as a basis for commercial product delivery.

1.2 Project Background

Software name: GOAT_666 Online Travel Platform.

Project source: the third course assignment (software requirements analysis). The project team has 13 members. Five members in roles A – E prepared, integrated, and reviewed this specification; development, interface, and testing members helped check the content.

Problem to be addressed: Travel information is scattered across several platforms, making it time-consuming for visitors to find destinations, attractions, guides, and hotel and ticket information. Booking processes are unclear, and users cannot easily experience the full process without the risk of real payment. The platform also lacks a place to retain community content such as travel notes, comments, ratings, and favorites.

The platform provides a complete process from finding travel information to sharing travel experiences. Visitors can browse and search destinations, attractions, guides, hotels, and tickets. Registered users can select a date and make a simulated booking, view or cancel orders, and publish travel notes, comments, ratings, and favorites. Administrators maintain platform information and user content. After the basic functions are complete, the platform plans to add personalized recommendations, content moderation, usage statistics, and AI trip suggestions.

1.3 Scope and Boundaries

Scope	Content
Initial release	Accounts (registration, login, logout, edit profile); browse and search travel information; view hotel and ticket products; simulated bookings, order lookup, cancellation, and simulated refunds; publish travel notes and add comments, ratings, and favorites; administrator maintenance of attractions, guides, products, and user content.
Extensions	Personalized recommendations (FR-10); content moderation and usage statistics (FR-11); AI trip assistant (FR-12).
Explicitly excluded	Real payments and refunds; offline redemption; multiple languages or currencies; native mobile apps (initial release is browser-only); third-party OTA integration.

Note: This platform is a teaching project. Bookings and refunds are simulated and do not involve actual funds. See Section 1.4 for related terms.

1.4 Terms and Abbreviations

Term or Abbreviation	Meaning	First Used
Visitor	A user who has not logged in. Can browse and search destinations, attractions, guides, hotels, and tickets, but cannot place orders or publish content.	2.2
Registered user	A regular user who has logged in. Can make simulated bookings, manage their own orders, and publish travel notes, comments, ratings, and favorites.	2.2
Administrator	A platform role responsible for maintaining attractions, guides, products, and user content.	2.2
Simulated booking	A booking process without real payment: after selecting a product, date, and quantity, the user places an order and the system creates an order and reserves inventory.	2.4
Simulated refund	A refund amount calculated and displayed under the cancellation rules; no real funds are involved.	2.4
Available quantity	The number of units of a product available for booking on a date (Inventory.availableCount).	3.1
Guide	An article about travel to a destination (Guide).	2.4
Travel note	An article about a trip published by a registered user (TravelNote).	2.4
SRS	Software Requirements Specification.	1.1
UML	Unified Modeling Language; used for the class diagram in this document.	3.1
FR NFR	Identifier prefixes for Functional Requirement Non-Functional Requirement.	3
SC	Identifier prefixes for Scenario	2.4
UI	UI Prototype	4
AT	Acceptance Test.	5

1.5 References

No.	Title and Version	Source or Link	Purpose
1	Third Course Assignment Requirements	https://bbs.csdn.net/topics/620549144	Grading criteria and delivery requirements
2	Project Definition (companion document)	Course materials	Functional boundaries and user flows
3	GB/T 9385-2008, Specification for Computer Software Requirements (current)	https://std.samr.gov.cn/gb/search/gbDetailed?id=71F772D7FB12D3A7E05397BE0A0AB82A	Principles for preparing the SRS
4	GB/T 45802-2025, Systems and Software Engineering—Life Cycle Processes—Requirements Engineering (current)	https://std.samr.gov.cn/gb/search/gbDetailed?id=36DE96AA3EACCD71E06397BE0A0A23D9	Requirements engineering process and expression principles
5	Specification Writing Assignments and Delivery Checklist	Course materials	Role assignments and delivery requirements

2. General Description

2.1 Product Goals and Application Scope

2.1.1 Product goal

Provide visitors with one place to look up destinations, attractions, guides, and hotel and ticket information; give registered users a risk-free simulated booking experience and community interaction; and allow administrators to maintain platform content.

2.1.2 Application scope

Accessible through a Web browser. Main users are students and general travelers interested in independent travel.

2.1.3 Product outlook

After the initial release completes the core process, add personalized recommendations, content moderation and usage statistics, and an AI trip assistant, forming a “search—book—travel—record” cycle.

2.2 User Roles

Role	Accessible Content	Main Tasks	Preconditions
Visitor	Browse destination, attraction, guide, and product lists and details	Find information; view prices and available quantities	No login required
Registered user	All visitor content plus simulated bookings, own orders, travel notes, comments, ratings, and favorites	Place and cancel orders, view simulated refunds, and publish interactive content	Logged in
Administrator	Platform administration area	Maintain attractions, guides, and products; handle user content	Administrator account

2.3 Operating Environment and Constraints

Operating environment: Modern Web browsers (latest versions of Chrome, Edge, and Firefox recommended), accessible on desktop and mobile browsers.

Technical constraints: The initial release is a Web application and does not connect to a real payment channel. Inventory is based on server data; quantities shown on pages are for display only. Dates use YYYY-MM-DD. Amounts are in Chinese yuan (¥), with two decimal places.

2.4 User Scenarios

All scenarios are described using “role and goal — trigger — preconditions — normal steps — exceptions or branches — outcome — related requirements.” Steps include only observable user and system behavior.

SC-01 Visitor Searches for Attraction Guides

Element	Content
Role and goal	A visitor wants to find attractions worth visiting and related guides for a destination.
Trigger	The visitor opens the platform home page.
Preconditions	No login required.
Normal flow	1. The visitor enters “Hangzhou” in the home-page search box; 2. The system returns results for Hangzhou destinations, attractions, and guides; 3. The visitor clicks the “West Lake” attraction to view its details; 4. The visitor opens a related guide from the detail page.
Exceptions and branches	If no results are found, display “No related content found. Try another keyword.” Visitors can view all information without logging in.
Outcome	The visitor obtains attraction and guide information and can view related products.
Related requirements	FR-02, FR-03, UI-01

SC-02 User Makes a Simulated Booking

Element	Content
Role and goal	A registered user selects a product after searching, completes a simulated booking, and generates an order.
Trigger	The user clicks a hotel or ticket product on an attraction detail page or product list.
Preconditions	The user is logged in; the product is “Listed”; the selected date has available inventory.
Normal flow	1. On the product detail page, the user selects the use date 2026-10-20 and quantity 2, then clicks [Simulated Booking]; 2. The system checks login status, whether the product is listed, and whether the available quantity for that date is at least 2; 3. After validation, the system deducts 2 from the available quantity for that date, generates an order with status “Reserved” and its line item, and calculates total amount = unit price × quantity; 4. The system opens the order confirmation page and displays the order number, details, and total; 5. The user can view the order under “My Orders.”
Exceptions and branches	Not logged in: open the login page and return to product details after login; Insufficient inventory: show “Insufficient inventory for the selected date. Adjust the quantity or date,” and do not create an order; Invalid quantity (0, negative, non-integer, or above the available quantity): show an error and block submission; Repeated click on [Confirm Order]: disable the button and use server-side idempotency so only one order is created; Product delisted: show “This product has been delisted.
Outcome	One order with status “Reserved” is created, the available quantity for that date decreases, and the page shows the order confirmation.
Related requirements	FR-04, FR-05, UI-02, UI-03, AT-F01–AT-F05

SC-03 Publish a Travel Note and Interact

Element	Content
Role and goal	A registered user publishes a travel note and receives comments, ratings, and favorites.
Trigger	The user clicks “Write a Travel Note” in the personal center.
Preconditions	The user is logged in.
Normal flow	1. Enter a title and body and upload images; 2. Click Publish and the travel note enters the public list; 3. Other users can comment, rate it (1–5 stars), and add it to favorites.
Exceptions and branches	Ask users who have not logged in to log in; show a message if an image exceeds the limit; hidden content is not displayed.
Outcome	The travel note is publicly visible and accumulates interaction data.
Related requirements	FR-07, FR-08, UI-05, UI-06

SC-04 Administrator Maintenance

Element	Content
Role and goal	An administrator maintains platform information and user content.
Trigger	The administrator enters the administration area after logging in.
Preconditions	Logged in with an administrator account.
Normal flow	1. Add or edit attractions, guides, and products; 2. Delist products that violate rules; 3. Hide travel notes or comments that violate rules.
Exceptions and branches	Show “Access denied” when a regular user tries to access the administration area.
Outcome	Platform content is maintained.
Related requirements	FR-09, FR-11, UI-07

SC-05 User Cancels an Order and Views a Simulated Refund

Element	Content
Role and goal	A registered user cancels their own “Reserved” order and sees the simulated refund.
Trigger	The user clicks [Cancel Order] on the “My Orders” page.
Preconditions	The order status is “Reserved,” and the user owns the order.
Normal flow	1. The system verifies ownership (rejecting another user's order) and the current status (only “Reserved” orders may be cancelled); 2. The system calculates a simulated refund under the cancellation rules: cancel at least 3 days before the use date for 100%, 1–2 days before for 50%, or on the use date for 0%; 3. The system changes the order status to “Cancelled,” records refundAmount, and restores the available quantity for the date by the cancelled quantity; 4. The page shows “Order cancelled. Simulated refund of ¥xx returned (simulated),” and updates the status in the order list.
Exceptions and branches	Repeated cancellation: show “Order already cancelled; no further action needed”; Completed order: cannot be cancelled; Unauthorized access to another user's order: show “Access denied” without disclosing order content.

Element	Content
Outcome	The order becomes “Cancelled,” inventory is restored, and the refund amount is displayed under the rules.
Related requirements	FR-06, UI-04, AT-F06–AT-F08

SC-06 Registered users use the AI travel assistant [Extension]

Element	Content
Role and goal	Registered users want to quickly generate a reference travel itinerary based on their destination.
Trigger	Registered users can click on the "AI Trip Assistant" entry in the navigation bar.
Preconditions	The user account is already logged in; the platform has the corresponding scenic spots and travel guide data for this destination.
Normal flow	1. The user accesses the AI itinerary assistant page (UI-08). 2. The user enters the destination name, sets the number of travel days, and can optionally fill in travel preferences (such as natural scenery, food exploration, family leisure). 3. The user clicks the "Generate Itinerary" button. 4. The system reads the attraction and guide data within the platform, generates a structured itinerary plan, and displays recommended attractions and simple travel tips by day. 5. The user browses the generated itinerary, and can view the jump to the detailed page of the attractions within the itinerary.
Exceptions and branches	1. AI service timeout: The page displays "The AI service is temporarily busy. Please try again later", and does not show invalid trips. 2. Destination lacking available platform data: It shows "This destination does not have sufficient reference materials. No itinerary can be generated. Please change the destination". 3. Visitors who are not logged in directly accessing the AI assistant page: Redirect to the login page, and then return to the AI assistant page after logging in. 4. User inputs an empty destination: It prompts "Destination is a required field. Please enter the destination".
Outcome	Normal: The page displays the itinerary reference plans generated by AI, categorized by days, and allows for navigation to the details of the corresponding attractions; Exception: A clear prompt is given, and no blank or garbled itinerary will be output.
Related requirements	FR-12、UI-08、AT-F10

3. Specific Requirements

3.1 Overall Class Diagram

Figure 3-1 is the overall class diagram. It covers core objects including users, destinations and attractions, guides, hotel and ticket products, inventory, orders, travel notes, comments, ratings, and favorites. Class names, states, and functional descriptions are consistent throughout the document.

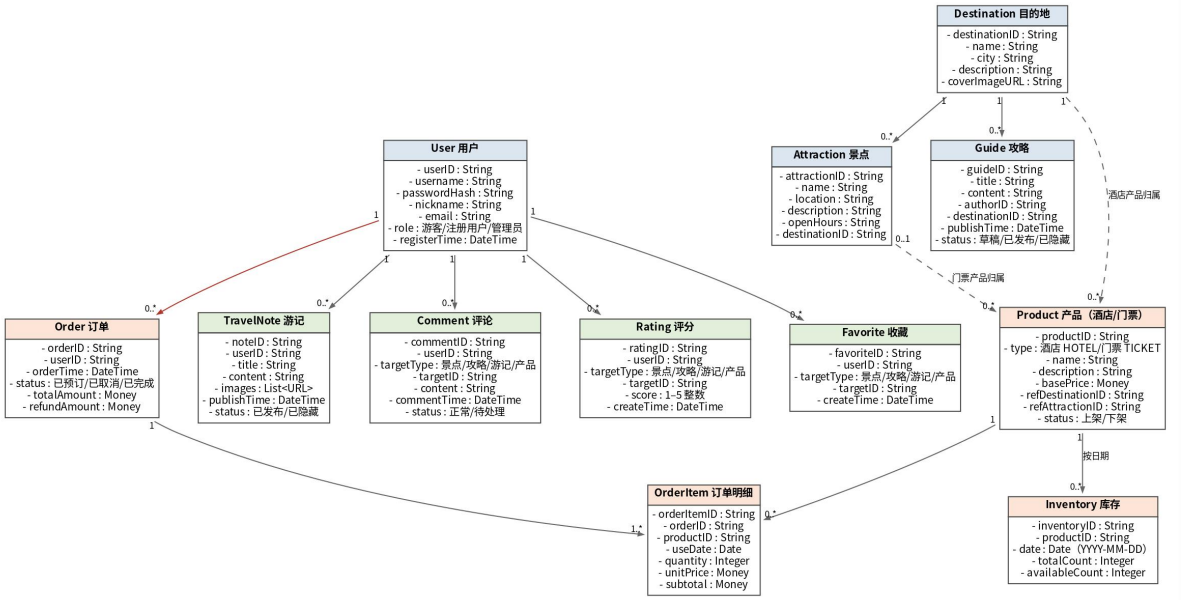


Figure 3-1 Overall Class Diagram

Class or Object	Key Attributes	Main Relationships and Multiplicity
User	userID, username, passwordHash, nickname, email, role (visitor/registered user/administrator), registerTime	1—0..* Order; 1—0..* TravelNote; 1—0..* Comment; 1—0..* Rating; 1—0..* Favorite
Destination	destinationID, name, city, description, coverImageURL	1—0..* Attraction; 1—0..* Guide; 1—0..* Product (hotel product association)
Attraction	attractionID, name, location, description, openHours, destinationID	Belongs to 1 Destination; 0..1—0..* Product (ticket product association)
Guide	guideID, title, content, authorID, destinationID, publishTime, status	Belongs to 1 Destination
Product	productID, type (hotel/ticket), name, description, basePrice, refDestinationID, refAttractionID, status (listed/delisted)	1—0..* Inventory; 1—0..* OrderItem
Inventory	inventoryID, productID, date, totalCount, availableCount	Belongs to 1 Product; productID + date is unique
Order	orderID, userID, orderTime, status (reserved/cancelled/completed), totalAmount, refundAmount	Belongs to 1 User; 1—1..* OrderItem
OrderItem	orderItemID, orderID, productID, useDate, quantity, unitPrice, subtotal	Belongs to 1 Order; references 1 Product
TravelNote	noteID, userID, title, content, images, publishTime, status	Belongs to 1 User
Comment	commentID, userID, targetType, targetID, content, commentTime, status	Belongs to 1 User; polymorphic target (attraction/guide/travel note/product)
Rating	ratingID, userID, targetType, targetID, score (1–5), createTime	Belongs to 1 User; polymorphic target
Favorite	favoriteID, userID, targetType, targetID, createTime	Belongs to 1 User; polymorphic target

Figure 3-1 Relationship notes

Orders and users: A registered user may have 0..* orders. Each order must belong to exactly one user (Order contains userID). Visitors must log in before placing an order, so every order is linked to a registered user.

Products and inventory: A product has multiple inventory records by date (Inventory, with a unique productID + date). On ordering, the system checks and deducts availableCount for the date; on cancellation, it restores the count. The quantity shown on the page reflects server data.

Orders and order items: An order contains 1..* OrderItems. An item records the product, use date, unit price, quantity, and subtotal. The order total equals the sum of item subtotals.

Destination — attraction — product: A destination contains 0..* attractions. Hotel products belong to a destination (refDestinationID); ticket products belong to an attraction (refAttractionID). This relationship determines where products appear in lists and details.

Community objects: Comment, Rating, and Favorite use targetType + targetID to refer to an attraction, guide, travel note, or product (polymorphic association). Each user may have at most one rating and one favorite for a given target.

3.2 Usability Requirements

ID	Requirement	Verification Method
NFR-U01	All form input errors must show a specific message and allow correction without clearing entered content.	Enter invalid values one by one; check messages and retention of entered content
NFR-U02	States such as no results, sold out, and no orders must have clear empty-state text and guidance.	Create empty-data scenarios and inspect each page
NFR-U03	Main actions (place order, cancel) must provide page feedback within 2 seconds.	Timed check
NFR-U04	Pages must be usable at 1440×900 and 375×667 viewport sizes.	Check key pages at both viewport sizes

3.3 Security and Access Requirements

ID	Requirement	Verification Method
NFR-S01	Regular users may view and manage only their own orders. Accessing another user's order returns “Access denied” without disclosing order information.	Use two accounts to access each other's order details
NFR-S02	When a visitor attempts a restricted action such as ordering or publishing, redirect to the login page.	Click restricted actions without logging in
NFR-S03	Passwords must be stored securely (for example, salted hashes), and login state must use a session/token mechanism.	Code review and API tests
NFR-S04	Regular users must not access administration pages or APIs.	Access the administration area with a regular account
NFR-S05	Validate invalid inputs (excessive length, illegal characters, out-of-range quantity) on the server and return a clear error.	Test boundary inputs

3.4 Other Quality Constraints

ID	Requirement and Applicable Conditions	Verification Method
NFR-O01	Placing and cancelling an order must keep inventory and order data consistent (transaction); concurrent orders must not oversell.	Load-test available quantities for the same date with concurrent orders
NFR-O02	Changes to order status must have audit records (operator, time, previous and new status).	Check administration logs or data
NFR-O03	The initial release must function consistently in major browsers (Chrome/Edge/Firefox).	Regression tests in all three browsers

4. Interface Prototypes

4.1 Page List and Navigation

Page ID	Page Name	Roles	Entry and Navigation	Related Scenario
UI-01	Home and Search	Visitor/registered user	Site navigation; search → results → details	SC-01
UI-02	Product Detail Page	Visitor/registered user	From list/details; [Simulated Booking] → UI-03; not logged in → login	SC-02
UI-03	Order Confirmation Page	Registered user	After ordering from details; confirm → UI-04; return → UI-02	SC-02
UI-04	My Orders Page	Registered user	Personal center → My Orders; [Cancel Order] → status update	SC-02 SC-05
UI-05	Travel Notes and Interaction Page	Visitor/registered user	Home/personal center; publish → travel note editor	SC-03
UI-06	Personal Center	Registered user	“My Account” in top bar; profile, my travel notes, my orders, my favorites	SC-03
UI-07	Administration Page	Administrator	Visible after administrator login	SC-04
UI-08	AI Trip Assistant Page (extension)	Registered user	To be added in the extension phase	SC-06

4.2 Key Page Prototypes

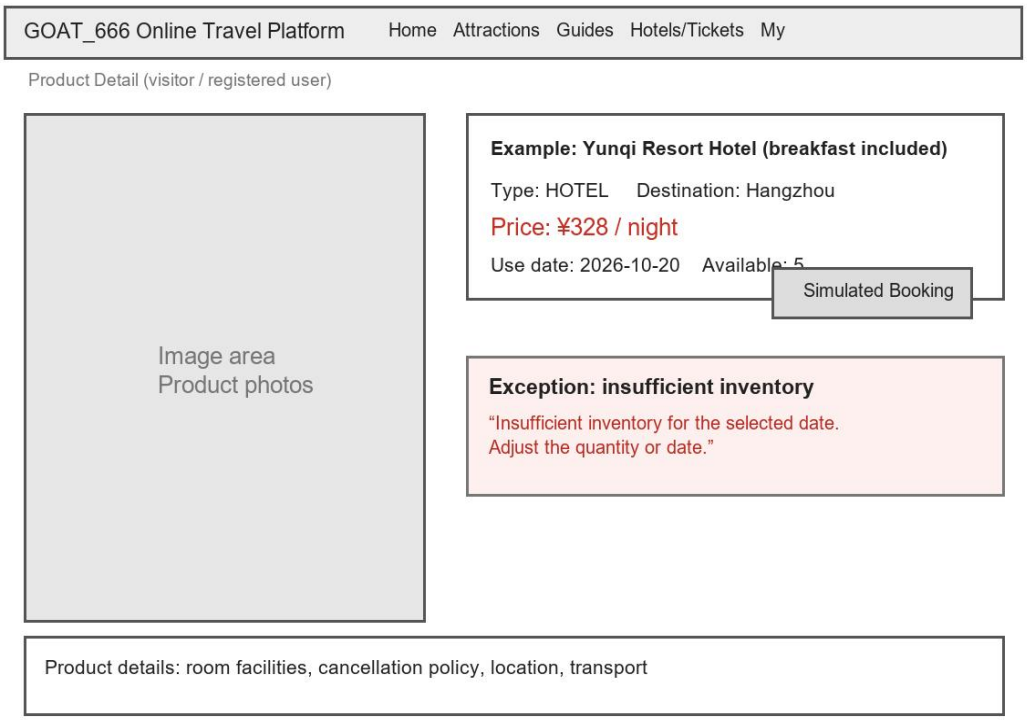


Figure 4-1 Product Detail Page Wireframe (UI-02)

Item	Description
Page ID and Role	UI-02; visitor/registered user
Main Controls and Inputs	Use-date selector (YYYY-MM-DD; only bookable dates), quantity selector (positive integer from 1 to available quantity), [Simulated Booking] button
Action and Navigation	Click [Simulated Booking] → validate → Order Confirmation Page UI-03; not logged in → login
Empty States and Errors	Disable sold-out dates; show insufficient inventory/invalid quantity errors; show a message for delisted products
Related Requirements	SC-02, FR-04, FR-05, AT-F01–AT-F05

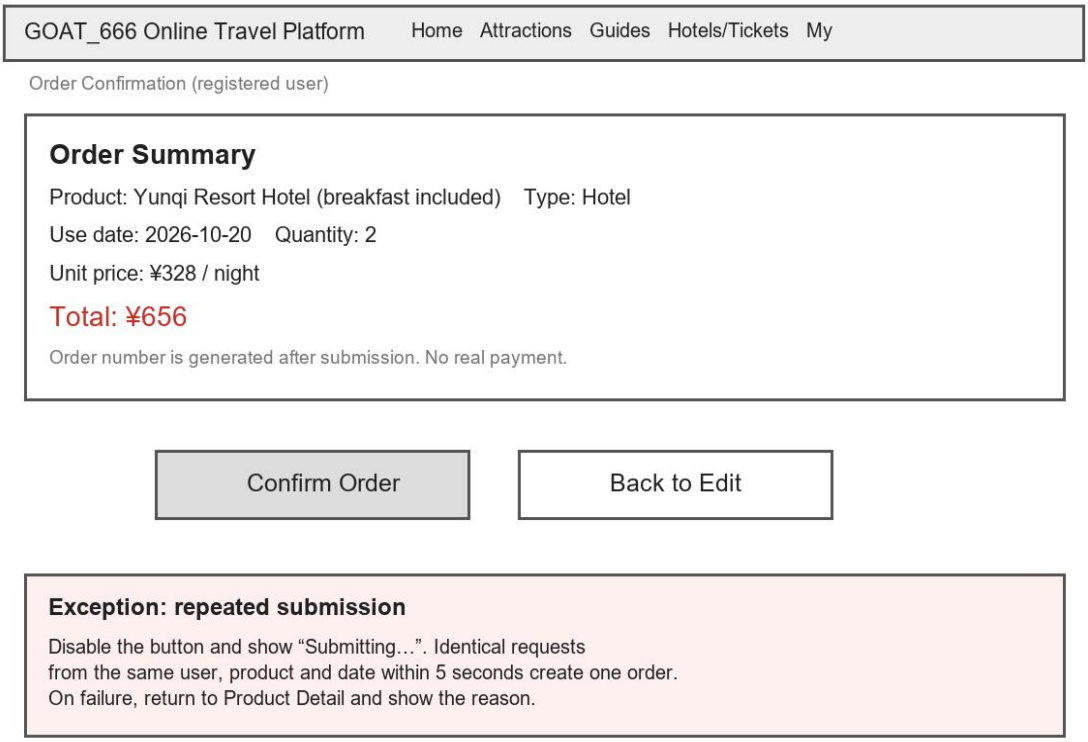


Figure 4-2 Order Confirmation Page Wireframe (UI-03)

Item	Description
Page ID and Role	UI-03; registered user
Main Controls and Inputs	Order summary (product, date, quantity, unit price, total), [Confirm Order], [Back to Edit]
Action and Navigation	Confirm → create order → My Orders Page UI-04; return → Product Detail Page UI-02
Empty States and Errors	Repeated submission: disable button and show "Submitting…"; server-side idempotency creates only one order; on failure return to UI-02 and show the reason
Related Requirements	SC-02, FR-05, AT-F02–AT-F05

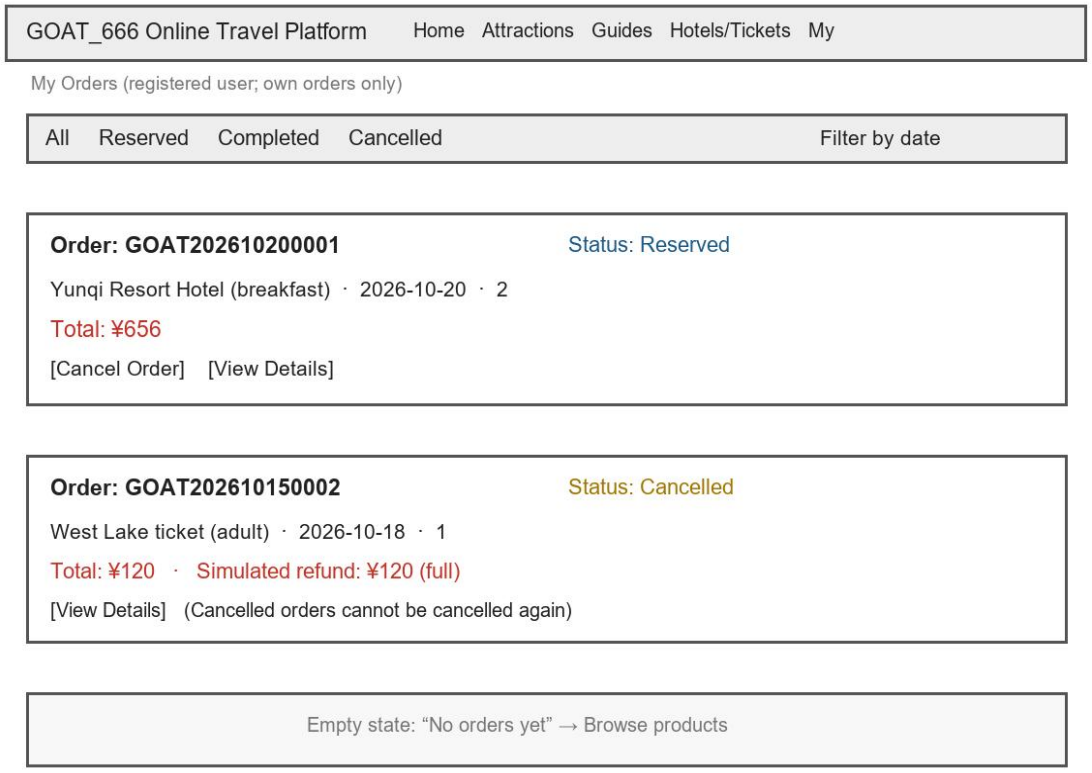


Figure 4-3 My Orders Page Wireframe (UI-04)

Item	Description
Page ID and Role	UI-04; registered user (own orders only)
Main Controls and Inputs	Status filter (all/reserved/completed/cancelled), date range filter, order cards, [Cancel Order], [View Details]
Action and Navigation	Cancel → status becomes “Cancelled” and simulated refund amount appears; details → Order Detail Page
Empty States and Errors	No orders: show “No orders yet” and guide user to browse products; cancelled orders cannot be cancelled again; unauthorized access to another user's order shows access denied
Related Requirements	SC-05, FR-06, AT-F06–AT-F08

5. Functional Description and Acceptance Verification

5.1 Detailed Functional Description

Table 5-1 lists the functional requirement identifiers.

ID	Name	Phase	Lead Role
FR-01	Account Management	Initial release	D
FR-02	Browsing Travel Information	Initial release	A
FR-03	Search	Initial release	A
FR-04	Hotel and Ticket Product Search	Initial release	B
FR-05	Simulated Booking (Place Order)	Initial release	B
FR-06	Order Management (Lookup, Cancellation, Simulated Refund)	Initial release	B
FR-07	Travel Notes	Initial release	C
FR-08	Comments, Ratings, Favorites	Initial release	C
FR-09	Platform Management	Initial release	D
FR-10	Personalized Recommendations	Extension	A
FR-11	Content Moderation and Usage Statistics	Extension	D
FR-12	AI Trip Assistant	Extension	C

Table 5-1 Functional Requirement Overview

FR-01 Account Management

Field	Function FR-01 Account Management
Phase and Roles	Initial release; visitors (register), registered users (log in/log out/edit profile), administrators (assign roles).
Inputs and Format	Username (unique), password (at least 8 characters including letters and numbers), email; profile fields are optional when editing.
Processing Rules	Notify on duplicate usernames; validate password format; manage login sessions; administrator roles are assigned in the administration area.
Outputs and State Changes	Go to the home page after successful registration/login; profile changes take effect immediately.
Exceptions and Access	Provide clear messages for duplicate usernames, invalid password format, and login failures.
Related Items	SC-02, UI-06

FR-02 Browsing Travel Information

Field	Function FR-02 Browsing Travel Information
Phase and Roles	Initial release; visitors/registered users.
Inputs and Format	Browse paginated destination, attraction, and guide lists; detail pages show text and images.
Processing Rules	Destination details show its attractions and guides; attraction/guide details show related products and content.
Outputs and State Changes	List and detail pages; show an empty state when there is no data.
Exceptions and Access	No access restrictions.
Related Items	SC-01, UI-01

FR-03 Search

Field	Function FR-03 Search
Phase and Roles	Initial release; visitors/registered users.
Inputs and Format	Keyword; optional filter (destination/attraction/guide).
Processing Rules	Fuzzy-match titles and summaries by keyword; paginate results; link results to detail pages.
Outputs and State Changes	Results list; show an empty state if there are no results.
Exceptions and Access	Do not search with an empty keyword; show a message.
Related Items	SC-01, UI-01

FR-04 Hotel and Ticket Product Search

Field	Function FR-04 Hotel and Ticket Product Search
Phase and Roles	Initial release; visitors and registered users may use it without logging in.
Preconditions	Product status is “Listed.”
Inputs and Format	Optional filters: destination/attraction (dropdown), product type (hotel/ticket), use date (YYYY-MM-DD). Sort: default/price ascending/price descending. List has 10 items per page.
Processing Rules	1. Return listed products matching the filters; 2. Show name, type, lowest price, and available quantity for the selected date in the list; 3. Show dates with availableCount = 0 as “Sold out” and make them unselectable; 4. Show description, bookable dates, price, and daily available quantities on the detail page; 5. Use real-time server data for available quantities.
Outputs and State Changes	Product list/detail pages; if no results, show “No products match the criteria. Adjust the filters.”
Exceptions and Access	If a product is delisted, show “This product has been delisted” on the detail page. No access restrictions.
Related Items	SC-01, SC-02, UI-02, AT-F01

FR-05 Simulated Booking (Place Order)

Field	Function FR-05 Simulated Booking
Phase and Roles	Initial release; registered users place orders (visitors must log in first).
Preconditions	User logged in; product listed; available quantity for the selected date is at least the requested quantity.
Inputs and Format	Product (product ID); use date (YYYY-MM-DD, bookable date only); quantity (positive integer, 1 ≤ quantity ≤ available quantity for that date, default 1).
Processing Rules	1. Check available quantity for the product + date in real time when placing an order; reject if insufficient; 2. In one transaction after validation, deduct the available quantity → create an order with status "Reserved" → create the order item; total = unit price × quantity; 3. Prevent duplicate submission: disable the button and show "Submitting…" after a click. The server handles identical requests from the same user for the same product and date within 5 seconds idempotently, creating only one order; 4. This is only a simulated booking, with no real payment.
Outputs and State Changes	Open the Order Confirmation Page (UI-03) showing order number, details, and total; order status is "Reserved"; available quantity for the date decreases.
Exceptions and Access	Not logged in → login page; insufficient inventory → "Insufficient inventory for the selected date. Adjust the quantity or date"; invalid quantity (0, negative, non-integer, above available amount) → show an error and block submission; delisted product → show a message; duplicate submission → create only one order and show "Order already submitted. Do not submit again."
Related Items	SC-02, UI-02, UI-03, AT-F02–AT-F05

FR-06 Order Management

Field	Function FR-06 Order Management
Phase and Roles	Initial release; registered users (own orders only). Administrators may view aggregate order data in moderation and statistics (without changing orders).
Preconditions	User logged in; an order belonging to the user exists.
Inputs and Format	Query filters: status (all/reserved/completed/cancelled), optional time range, optional exact order number.
Processing Rules	1. Return only the user's orders, sorted by order time descending by default; 2. Only "Reserved" orders may be cancelled; 3. Calculate simulated refunds: at least 3 days before use date, 100%; 1–2 days before, 50%; on the same day, 0%; 4. After successful cancellation, change the status to "Cancelled," record refundAmount, and restore the available quantity for that date by the cancelled quantity; 5. "Completed" and "Cancelled" orders cannot be cancelled again. For a repeat cancellation, show "Order already cancelled; no further action needed."
Outputs and State Changes	Order list with status labels; order details with items and refund amount; after successful cancellation, show "Order cancelled. Simulated refund of ¥xx returned (simulated)."

Exceptions and Access	Another user's order → show “Access denied” without displaying content completed/cancelled order → show a cannot-cancel message nonexistent order → show a message.
Related Items	SC-05, UI-04, AT-F06–AT-F08

Order state transitions are shown in Figure 5-1:

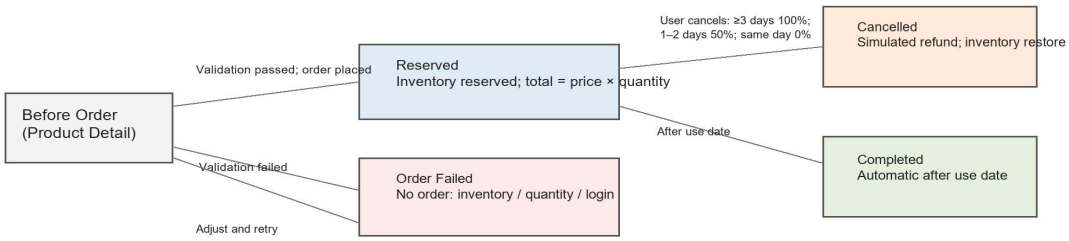


Figure 5-1 Order State Transition Diagram

State notes: If validation fails before an order is placed (insufficient inventory, invalid quantity, or not logged in), no order is generated. A successful order enters “Reserved” (inventory has been reserved). User cancellation changes it to “Cancelled” (a simulated refund is calculated and inventory restored). After the use date, the system automatically marks it “Completed.”

FR-07 Travel Notes

Field	Function FR-07 Travel Notes
Phase and Roles	Initial release; registered users.
Inputs and Format	Title (required, at most 50 characters), body (required), images (at most 5 MB each, up to 9).
Processing Rules	Authors may publish, edit, and delete their own travel notes. Hidden content is not shown publicly.
Outputs and State Changes	A published note appears in the public list; edits/deletions update the list.
Exceptions and Access	Users who have not logged in cannot publish; show a message if image limits are exceeded.
Related Items	SC-03, UI-05, UI-06

FR-08 Comments, Ratings, Favorites

Field	Function FR-08 Comments, Ratings, Favorites
Phase and Roles	Initial release; registered users.
Inputs and Format	Comment (at most 500 characters); rating of 1–5 stars; favorite target (attraction/guide/travel note/product).
Processing Rules	Each user may have at most one rating and one favorite for a target (which may be changed or removed); content moderation affects comment visibility.
Outputs and State Changes	Interaction data appears on the target detail page.
Exceptions and Access	Users who have not logged in cannot act; show a message for a deleted target.
Related Items	SC-03

FR-09 Platform Management

Field	Function FR-09 Platform Management
Phase and Roles	Initial release; administrators.
Inputs and Format	Add/edit/delete forms for attractions, guides, and products; listed/delisted toggle.
Processing Rules	Administrators maintain attractions, guides, and products, and may hide travel notes and comments that violate rules.
Outputs and State Changes	Front-end content updates following administration changes.
Exceptions and Access	Show “Access denied” if a non-administrator accesses the administration area.
Related Items	SC-04, UI-07

FR-10 Personalized Recommendations

Field	Function FR-10 Personalized Recommendations
Phase and Roles	Extension; registered users (new users see popular content).
Inputs and Format	User favorites and content popularity data (internal to the system).
Processing Rules	Recommend attractions and guides based on favorites and popularity; show popular content to new users.
Outputs and State Changes	Display in the home-page recommendation area.
Exceptions and Access	Fall back to popular content if data is insufficient.
Related Items	UI-01

FR-11 Content Moderation and Usage Statistics

Field	Function FR-11 Content Moderation and Usage Statistics
Phase and Roles	Extension; administrators.
Inputs and Format	Comments marked for handling; statistics time range.
Processing Rules	Flag comments for administrator action; show basic usage and order statistics.
Outputs and State Changes	Display on the administration statistics page.
Exceptions and Access	Not visible to non-administrators.
Related Items	UI-07

FR-12 AI Trip Assistant

Field	Function FR-12 AI Trip Assistant (draft, extension)
Phase and Roles	Extension; registered users.
Inputs and Format	Destination (required), number of days, optional preferences.
Processing Rules	Generate trip suggestions based on platform content. The AI does not create or change orders directly.
Outputs and State Changes	Trip suggestion page.
Exceptions and Access	Show a clear message on service timeout or when no usable material is available.
Related Items	UI-08

5.2 Input and Output Formats

Object/Action	Input Fields and Constraints	Successful Output	Failure or Empty Result
Product search	Type (hotel/ticket, optional); destination/attraction (optional); date (YYYY-MM-DD, optional); sort (default/price ascending/price descending)	Product list (up to 10 per page): name, type, lowest price, available quantity	No results: “No products match the criteria. Adjust the filters.”
Product details	Product ID; optional date	Name, description, price, bookable dates, daily available quantity	Missing/delisted product: “This product is unavailable or does not exist.”
Simulated order	Product ID; use date (YYYY-MM-DD, bookable); quantity (positive integer from 1 to available quantity)	Order number, items (product/date/quantity/unit price/subtotal), total, status “Reserved”	Not logged in → login; insufficient inventory/invalid quantity/delisted → corresponding message, no order created
Order lookup	Status (all/reserved/completed/cancelled); optional time range; optional order number	Own order list (order number, product, date, quantity, amount, status)	No orders: “No orders yet,” with guidance to browse products
Cancel order	Order ID (own orders only)	Status “Cancelled,” simulated refund amount, available inventory restored	Repeated cancellation: “Order already cancelled; no further action needed”; completed order cannot be cancelled; unauthorized: “Access denied”
Refund rules	Difference in days d between cancellation time and use date (calculated by system)	$d \geq 3$: 100% refund; $1 \leq d < 3$: 50%; $d < 1$ (same day): 0%	—

5.3 Interface Acceptance Criteria

Acceptance ID	Page/Control	Action and Conditions	Expected Result	Prototype
AT-UI01	Home search box	Enter a keyword and click Search	Go to results; show an empty state if none	UI-01
AT-UI02	Product Detail Page	Select date and quantity and click [Simulated Booking]	Go to order confirmation after validation	UI-02
AT-UI03	[Confirm Order] on Order Confirmation Page	Click again immediately after the first click	Disable button and show “Submitting…”; create only one order	UI-03
AT-UI04	My Orders Page	View list after cancelling an order	Status updates to “Cancelled,” refund amount displayed	UI-04
AT-UI05	Administration Page	Regular user accesses it	Show “Access denied”	UI-07
AT-UI06	Travel Notes/Personal Center/AI Pages	Check each page when its prototype is complete	Fill in after role C completes the prototypes	UI-05/06/08

5.4 Functional Acceptance Criteria

Test ID	Related Req.	Initial Data and Steps	Expected Result
AT-F01	FR-04	Listed hotel “Yunqi Resort Hotel” has 5 units available for 2026-10-20; a visitor opens the product list and filters by that date	The list shows the product, quantity 5, and status “Bookable”, an empty-result scenario shows an empty state
AT-F02	FR-05	User logged in; product quantity 5; select 2026-10-20 and quantity 2; click [Confirm Order]	Create an order with status “Reserved”; total = unit price × 2; quantity becomes 3; order confirmation shows the order number
AT-F03	FR-05	Product quantity 1; submit quantity 2	Reject the order; show “Insufficient inventory for the selected date”; do not create an order; quantity unchanged
AT-F04	FR-05	Submit quantity 0 -1 2.5	Show “Quantity must be an integer of at least 1”; block submission
AT-F05	FR-05	Click [Confirm Order] twice rapidly (less than 5 seconds apart)	Create only one order; on the second click show “Order already submitted. Do not submit again.”
AT-F06	FR-06	For a reserved order at least 3 days before use date, click [Cancel Order]	Status “Cancelled”; 100% simulated refund; available quantity restored
AT-F07	FR-06	Cancel one reserved order 1 day before use date and another on the use date	Show 50% and 0% simulated refunds, respectively; both statuses “Cancelled”
AT-F08	FR-06	User A opens user B's order details directly; cancel an already cancelled order again	Show “Access denied” without order content; for repeated cancellation show “Order already cancelled; no further action needed”
AT-F09	FR-01–FR-03	Smoke tests for registration, search, and information browsing	Cover the SC-06 AI trip assistant extension scenario.
AT-F10	FR-07–FR-12	Community and extension function tests	Cover the SC-06 AI trip assistant extension scenario.

6. Process Records and Revisions

6.1 Design Decisions, Issues, and Solutions

Issue ID	Issue Encountered	Options Considered	Chosen Approach and Reason	Status
ISS-01	How should the simulated refund amount be defined?	Option 1: full refund in all cases; Option 2: tiered refund by days before use date	Choose Option 2 (at least 3 days: 100%; 1–2 days: 50%; same day: 0%). It better reflects cancellation costs in real booking scenarios, and the rule can be verified directly in acceptance tests.	Decided
ISS-02	When should inventory be deducted, and how can overselling be avoided?	Option 1: deduct in the transaction when the order is submitted; Option 2: reserve inventory and release it after a timeout; Option 3: deduct after payment (not applicable to simulated booking)	Choose Option 1: check quantity → deduct → create order in one database transaction; disable the front-end button and use server-side idempotency to prevent duplicates. Reserve the real-payment option for an extension.	Decided
ISS-03	Repeated clicks on Submit create duplicate orders	Option 1: disable button only on front end; Option 2: disable button plus server-side idempotency	Choose Option 2: disabling the front end prevents ordinary repeated clicks, while server-side idempotency for identical requests by the same user, product, and date within 5 seconds ensures data consistency.	Decided
ISS-04	How to model “product ownership” in the overall class diagram	Associate hotel products with destinations and ticket products with attractions, or associate both with a general “content object”	Choose the former (refDestinationID refAttractionID), consistent with product list filters; development team to verify.	Pending review
ISS-05	Inconsistent identifiers and terms when merging multiple authors' work	Each author uses their own, or agree on an identifier system and glossary first	Choose the latter: establish FR/SC/UI/AT identifiers and the glossary first, then role A checks consistency during integration.	Decided

6.2 User Validation Record

People and Date	Validation Task	Feedback or Issue	Corresponding Change/Plan
Planned interviews with 2–3 target users (date to be set)	Review SC-02/SC-05 scenarios and product detail and order page wireframes	Awaiting feedback	Revise scenarios and prototypes based on feedback, then update this document

Note: User validation has not yet been conducted. This is currently a plan, not a completed activity. Update this table after it takes place.

6.3 Unfinished Items and Follow-up Updates

Item	Current Status and Reason	Owner	Due Date	Update Location
Home search, travel notes, personal center, administration, and AI page wireframes	Awaiting compilation by role C; no wireframes yet	C	Before 2026-10-18	4.2
Detailed acceptance tests for FR-01–03 and FR-07–12	Awaiting module acceptance points from roles A/C/D	E	Before 2026-10-18	5.4
High-fidelity prototypes	Wireframes used instead	C	Extension phase	4.2
Real user interviews	Planned	Whole team	Before 2026-10-18	6.2
Development review of overall class diagram	Backend development members must confirm transaction and inventory approach	B + development	Before 2026-10-18	3.1

6.4 Requirements Traceability and Consistency Check

Scenario	Functional Requirements	Class Diagram Objects	Prototype	Acceptance Test
SC-01	FR-02, FR-03	Destination, Attraction, Guide	UI-01	AT-F09
SC-02	FR-04, FR-05	Product, Inventory, Order, OrderItem	UI-02 UI-03	AT-F01– AT-F05
SC-03	FR-07, FR-08	TravelNote, Comment, Rating, Favorite	UI-05 UI-06	AT-F10
SC-04	FR-09, FR-11	User (administrator), Comment, Product	UI-07	AT-F10
SC-05	FR-06	Order, Inventory, OrderItem	UI-04	AT-F06– AT-F08
SC-06	FR-10–FR-12	User, Destination, Attraction, Guide	UI-08	AT-F11